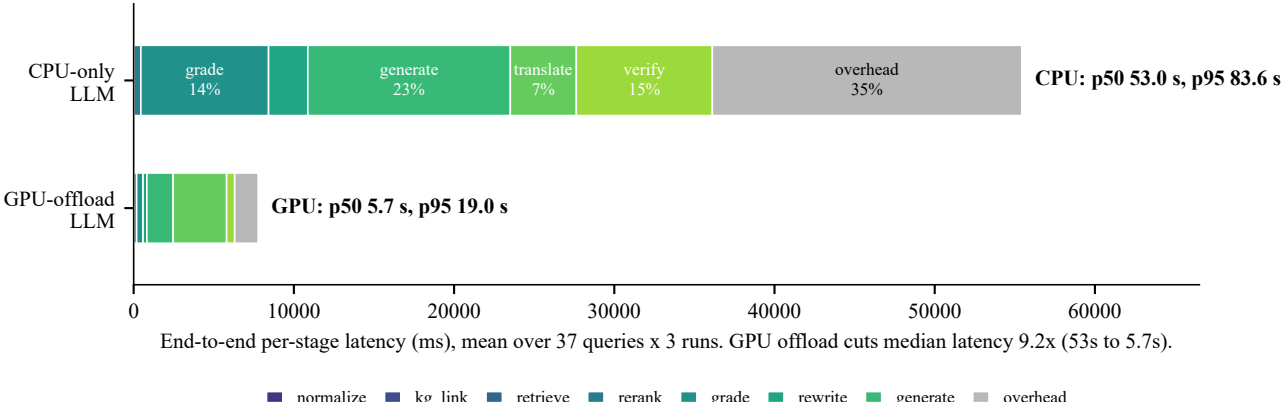
CPU-only LLM
grade 14%
generate 23%
translate 7%
verify 15%
overhead 35%
CPU: p50 53.0 s, p95 83.6 s
GPU-offload LLM
GPU: p50 5.7 s, p95 19.0 s
0
10000
20000
30000
40000
50000
60000
End-to-end per-stage latency (ms), mean over 37 queries x 3 runs. GPU offload cuts median latency 9.2x (53s to 5.7s).
normalize
kg_link
retrieve
rerank
grade
rewrite
generate
overhead